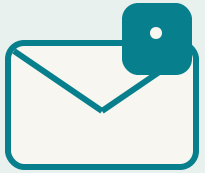
Why this design deserves credit.



Stratified randomisation

Helps balance sex and BMI across assigned groups.



Concealed allocation

Reduces scope for selection based on the next assignment.



Shared support

Makes unequal attention a less likely explanation.



Calibrated weighing

Less reliance on participants' own reports of their weight.

A calorie gap accompanied the result.

HKD

Energy restriction
+ carbohydrate limit

−704

kcal/day

ERD

Energy restriction

−423

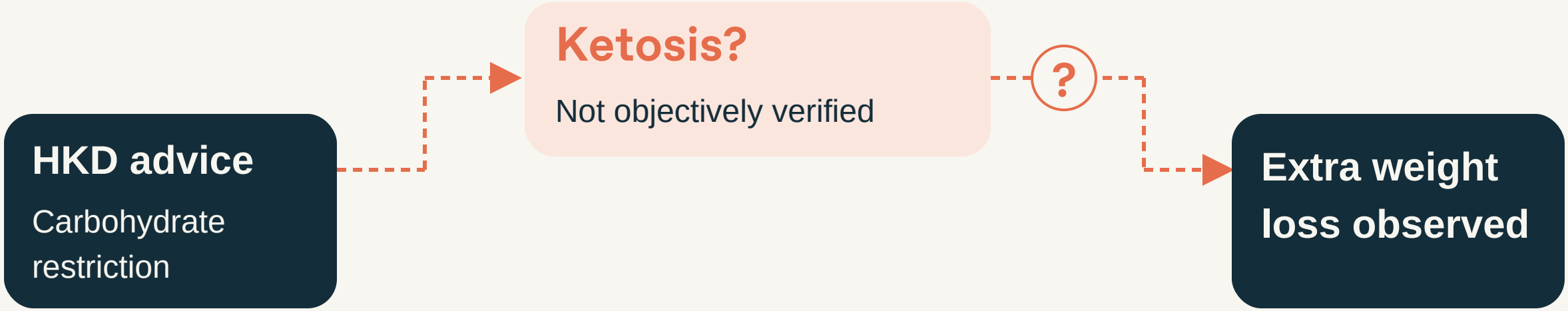
kcal/day

Calorie advice ≠ matched actual intake.

No controlled feeding ensured equal calorie intake.

Lim et al. 2024 · Methods 2.3, 2.5, 2.7 + Table 5 · Values are self-reported changes, not absolute intake.

Did ketosis explain the extra loss?



Dashed arrows = possible paths · ? = not established here.
Paths may overlap; their contributions were not isolated.

Lim et al. 2024 · Table 2, Table 5 + Discussion · Arrows do not represent estimated or demonstrated mediation.

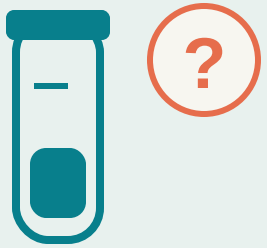
Calories offer another explanation.



Dashed arrows = possible paths · ? = not established here.
Paths may overlap; their contributions were not isolated.

Lim et al. 2024 · Table 2, Table 5 + Discussion · Arrows do not represent estimated or demonstrated mediation.

Two gaps. One mechanism limitation.



Unverified ketosis

Who reached it?
Was it sustained?



Calories not matched

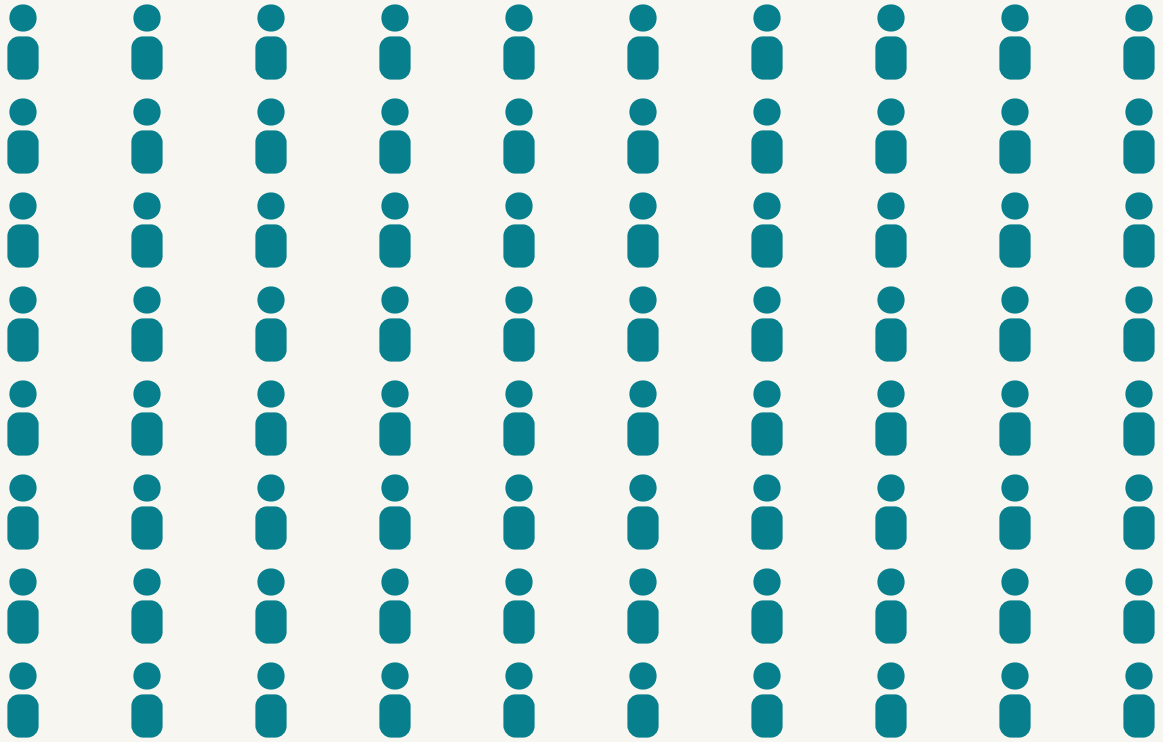
Would extra loss persist
at equal intake?

Two evidence gaps limit one mechanism claim.

Eating less can still be part of the programme’s benefit.

The programme comparison does not isolate a ketosis-specific effect independent of energy intake.

Who stays in the picture?



Observed Same 80-person cohort

80

randomized at baseline

100% in the starting cohort

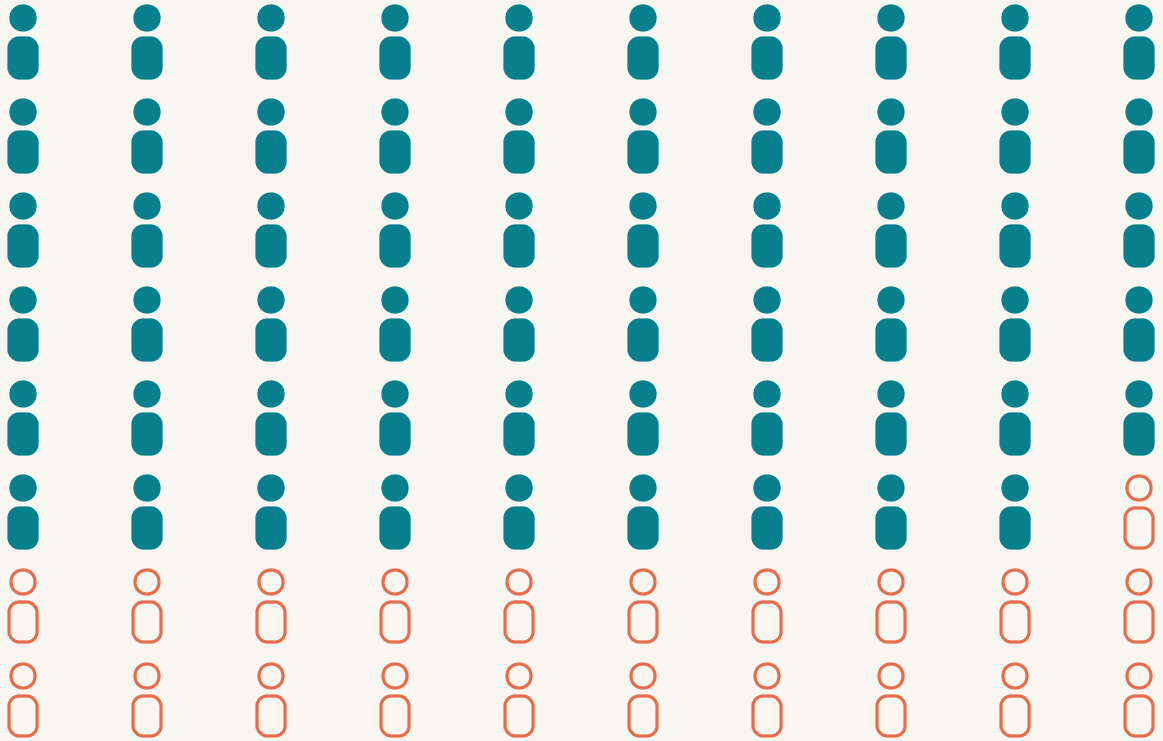
What happens next?

Follow the same cohort.

One figure = one participant

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

Who stays in the picture?



● Observed ○ Missing at 6 months

59

measured at 6 months

74% retained

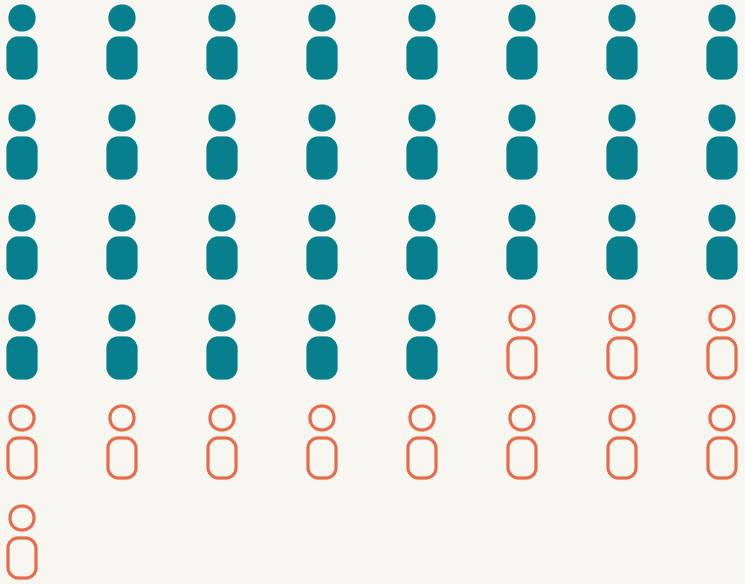
21 outcomes unknown

Missing weight ≠ failed diet.

HKD missing: 12 · ERD: 9

Lim et al. 2024 · Figure 1 + Table 2 · 6-month counts: HKD 29/41; ERD 30/39.

The missing weights could matter.



HKD: 29 measured + 12 missing

KNOWN

12 weights were not measured.

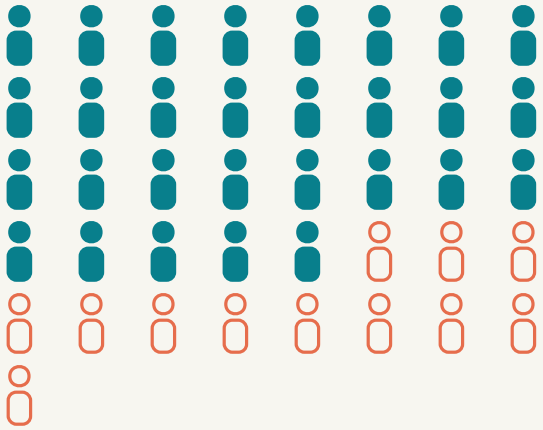
UNKNOWN

Less, similar or greater weight loss?

Their dietary adherence may be good.

Their unknown outcomes could change the group estimate.

Missing weights may bias the result.



29 measured
12 missing

IF POORER HKD RESPONDERS WERE MISSING…

The measured HKD subset would look more successful.

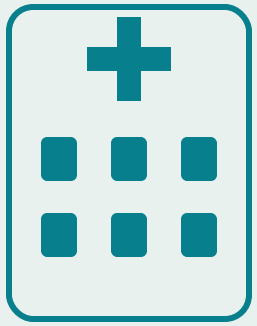
The apparent HKD advantage could be exaggerated.

Potential attrition bias; other missingness patterns could shrink the gap.

Missing-outcome handling is not clear in the report.

Figure 1 + Table 2 + Methods 2.9 · The actual direction of any missing-data bias cannot be established here.

Who does this sample represent?



One Singapore hospital

Healthcare staff

Single-centre recruitment

Women

86%

69 of 80 participants

Only 11 men



Access required

English literacy
+ smartphone

Selected volunteers

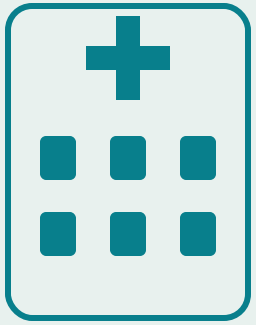
Some diseases excluded

Random assignment ≠ a representative population sample.

Lim et al. 2024 · Methods 2.2 + Table 1 · A limitation of generalisability, not proof that the within-study result is false.

Would it work the same elsewhere?

STUDIED SETTING



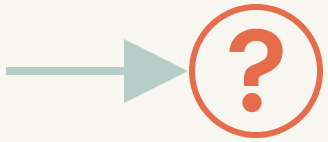
Seven workshops



App + dietitian coaching

Six-month support

Selected participants



Other adults?



Less support?

Same benefit is untested.

Limited generalisability ≠ evidence of no benefit elsewhere.

Lim et al. 2024 · Methods 2.5 · The arrow asks about transfer to other settings; it is not a biological causal claim.

Three limits. Three different questions.



WHY?

Mechanism was not isolated.

What produced the extra loss?



HOW RELIABLE?

Missing outcomes may affect the estimate.

Which outcomes are absent?



FOR WHOM?

Sample and conditions limit applicability.

Beyond similar people and support?

A useful programme comparison—with boundaries on the conclusions.

Lim SL et al. (2024), Nutrients 16:4380 · doi:10.3390/nu16244380